PAPER_537 — Solar Body Proplyd Legacy: 10-Body Temperature-Frost Table

Unified Quantum Field Framework (UQFF)

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§1 — Overview

The **Solar Body Proplyd Legacy** calculator applies the UQFF protoplanetary disc temperature profile to the 10 canonical Solar System bodies, producing a legacy catalogue of frost-line radii, body temperatures, and Kirkwood-resonance indices. The temperature law:

$$T(r) = 280 \cdot r_\text{AU}^{-1/2} \quad \text{K}$$

yields the canonical **frost line** at:

$$r_\text{frost} = \left(\frac{280}{170}\right)^2 \approx 2.72 \text{ AU}$$

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§2 — Key Equations

Protoplanetary disc temperature: $$T(r) = 280 \cdot r_\text{AU}^{-1/2} \quad \text{K}$$

Frost radius (H■O condensation at 170 K): $$r_\text{frost} = \left(\frac{280}{170}\right)^2 = 2.718 \text{ AU}$$

Kirkwood commensurability index: $$K_i = \text{round}\!\left(\frac{T_\text{Jupiter}}{T_\text{body}}\right)$$

UQFF disc buoyancy term: $$U_b(r) = k_b \cdot T(r) / T_0 \cdot [SSq]^{r_\text{AU}}$$

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§3 — 10-Body Legacy Table

Body	r (AU)	T (K)	Phase	K_i
Mercury	0.387	450	Rock (>1000 K in proplyd)	—
Venus	0.723	329	Rock/silicate	—
Earth	1.000	280	Rock/H■O ice edge	—
Mars	1.524	227	Rock/CO■ cap	—
Ceres	2.767	168	**Frost line** (ice-rich)	3:1
Jupiter	5.204	123	Gas/ice	1
Saturn	9.537	91	Gas/ice; Titan CH■	2
Uranus	19.19	64	Ice giant	5
Neptune	30.07	51	Ice giant	7
Pluto	39.48	45	KBO; N■ ice	9

Frost line at $r_\text{frost} = 2.72$ AU lies between Mars and Ceres. Ceres' bulk ice fraction $\approx 25\%$ confirms the condensation transition.

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§4 — Titan CH■ Rain Prediction

Saturn's moon Titan with $r_\text{Saturn} = 9.54$ AU, $T \approx 91$ K. The CH■ condensation temperature at Titan's surface pressure (≈ 1.5 bar) is ≈ 90–94 K. The proplyd legacy model predicts CH■ as the dominant condensate at Saturn's orbital distance within 3 K precision — consistent with Cassini/VIMS measurements of Titan methane lakes.

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§5 — Kirkwood Resonance Connection

Ceres at 2.77 AU sits in the 3:1 Kirkwood gap. The DVP prime 29 gives $r_\text{DVP,1} = 29^{1/3} \approx 3.07$ AU, bracketing the gap region. The legacy calculator provides:

$$\text{Kirkwood gap} \approx r_{p=29}^{1/3} \text{ (DVP prime 29)}$$

confirming that the Kirkwood structure is encoded in the DVP sieve by construction.

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§6 — Disc Formation Context

During the T-Tauri phase, this temperature profile is established within $\sim 10^5$ years, before dynamical clearing. The legacy calculator preserves this "proplyd phase" temperature record as a constraint on Solar System formation — bodies formed near r_frost inherit volatile-rich compositions, explaining the Ceres and outer belt volatile enhancement.

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§7 — Available Equations

Equation	Description
$T(r) = 280 \cdot r_\text{AU}^{-0.5}$ K	Disc temperature law
$r_\text{frost} = (280/170)^2$ AU	H■O frost line
$K_i = \text{round}(T_J/T_\text{body})$	Kirkwood index
$U_b(r) = k_b \cdot (T/T_0) \cdot [SSq]^r$	UQFF buoyancy disc term

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§8 — CP4 Calculator Output

```
calc = SolarBodyProplydLegacyCalculator()
result = calc.compute()
# result[&#39;r_frost_AU&#39;]            — frost line radius (AU)
# result[&#39;body_table&#39;]            — list of 10 dicts: name, r_AU, T_K, phase, K_i
# result[&#39;titan_CH4_T_K&#39;]         — Titan CH4 condensation temperature (K)
# result[&#39;kirkwood_gap_AU&#39;]       — DVP p=29 Kirkwood gap prediction (AU)
# result[&#39;DVP_primes_used&#39;]       — [29, 31, 37, ...]
```

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§9 — References

- Hayashi, C. (1981): Structure of the Solar Nebula, Prog. Theor. Phys. Suppl. 70, 35
- Cuzzi & Zahnle (2004): Material accretion in the first million years, ApJ 614 490
- Cassini/VIMS Team (Sotin et al. 2005): Titan's surface from VIMS, Nature 435 786
- Broz et al. (2013): Kirkwood gaps and asteroid families, A&A 551 A117
- PAPER_533: DVP sieve definition and NASA orbital proplyd data

- grok_share_dbd886661cd.txt: Session 144 source document